Space And Labor Markets

Urban Access, Commuting, Amenities, And Local Opportunity

Special Topic 4: Urban and Labor – Week 1

Central question

- How does space enter labor-market choice, wage determination, and worker welfare?
- Workers choose over job-residence bundles, not jobs from a placeless menu.
- Wages, rents, commuting costs, amenities, risks, and outside options jointly define opportunity.
- Week 1 discipline: separate workplace, residence, access, wages, rents, and welfare.

Why this is labor economics

- Space changes which jobs workers can search for, reach, accept, and keep.
- Local labor markets are shaped by commuting and search frictions, not only by administrative borders.
- A high nominal wage can be offset by rent, travel time, congestion, risk, or weak outside options.
- Urban wage premia raise labor questions: productivity, learning, sorting, or compensation?

Worker job-residence choice

$$U_{ijr} = w_j - R_r - \tau(d_{jr}) + A_r + \varepsilon_{ijr}$$

Term	Interpretation
w_j	earnings at workplace j
R_r	housing cost at residence r
$\tau(d_{jr})$	generalized commuting cost between j and r
A_r	amenity or disamenity value at residence r
ε_{ijr}	worker-specific fit with the bundle

The city as a labor-market bundle

Object	Labor meaning
Workplace wage	pay attached to a job location
Residential rent	cost of accessing a place while working
Commuting cost	time, money, reliability, effort, and risk
Amenities	local nonwage welfare components
Job access	jobs reachable within a travel-cost budget
Outside option	best connected alternative nearby or elsewhere

- The city is not a label. It is a set of constraints and opportunities.

Local labor markets and spatial search

$$\mathcal{J}_i(c) = \{j : \tau(d_{ij}) \leq c\}$$

- Accessible jobs decline as distance, time, cost, congestion, unreliability, or risk rises.
- Access is the feasible set; realized commute is the chosen match.
- Flow-based markets use commuting, travel time, and search links.
- Administrative markets are useful containers, but not automatically the economic market.

Residence and workplace are different objects

Measure	Main question
Residence-based wages	What do residents earn and face?
Workplace-based wages	What jobs and pay are located here?
Commuting flows	How are residences connected to workplaces?
Local rents	What cost is capitalized into residential access?
Travel-time markets	Which jobs are actually reachable?

- Conflating residence and workplace can misstate welfare, policy incidence, and local adjustment.

Commuting costs, rents, and amenities

- A job offer is valuable only if it can be paired with a feasible residence.
- A residence is valuable partly because of the jobs it gives access to.
- Commuting links labor demand to households without requiring migration.
- Amenities and disamenities enter welfare even when payroll data omit them.
- Housing and transportation constraints can make the same wage worth different amounts to different workers.

Spatial equilibrium and welfare

$$w_\ell - R_\ell + A_\ell = \bar{U}$$

- Wages, rents, amenities, and mobility jointly allocate workers across places.
- High wages need not imply high utility when rents or commuting costs are high.
- Low wages need not imply low utility when housing is cheap or amenities are valuable.
- When mobility is constrained, equilibrium gaps become labor-market objects.

Urban wage premium

$$\Delta w^{urban} = \Delta w^{productivity} + \Delta w^{learning} + \Delta w^{sorting} + \Delta w^{compensating}$$

Channel	Labor interpretation
Productivity	density raises output or match quality
Learning	workers accumulate skills faster in cities
Sorting	higher-productivity workers select into dense places
Compensation	wages offset rents, congestion, risk, or disamenities

Job access and the geography of work

- Jobs can move across space even when workers do not.
- Job suburbanization changes who can reach employment at acceptable commuting cost.
- Access shocks can affect employment without a simple wage-price story.
- The empirical object is not only distance; it is generalized travel cost, information, schedules, safety, and feasible job matches.

Methods discipline

- Separate residence-based and workplace-based measurement.
- Separate nominal wages from real opportunity.
- Separate access from realized commute.
- Separate spatial selection from causal local effects.
- Separate geographic boundaries from flow-based labor markets.

Week 1 lab

- Reproduce: compute job access in a deterministic synthetic city.
- Diagnose: classify access, commute, wage, rent, amenity, and equilibrium objects.
- Transfer: study a small commuting shock and ask who adjusts through commuting versus migration.
- Goal: turn a place-specific setting into a portable labor-market mechanism.

Bridge to Week 2

- Week 1 defines space as jobs, residences, commuting, rents, amenities, risks, and outside options.
- Week 2 asks when housing and moving costs block workers from reaching better labor-market opportunities.
- Central next question: who captures a local wage gain when rents and mobility constraints adjust?

Takeaways

- Local labor markets are access systems, not just places on a map.
- Workplace and residence must be separated before interpreting wages or welfare.
- Urban wage premia are labor-market objects with multiple possible channels.
- The rest of the course studies constraints that reshape the feasible set for work.